

Quantitative analysis of rim crest asymmetry in fresh lunar simple craters using Chandrayaan-2 TMC-2 DEM

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ABSTRACT

Impact cratering is the most common geological process in the solar system. Understanding the factors that control crater morphology and deriving crater parameters is essential for effectively using craters as analytical tools. Being pristine, lunar impact craters have been widely studied in recent years, with a focus on parameters such as diameter, depth, ejecta patterns, and the formation of the central peak. However, rim crest asymmetries, that is, differences in elevation, shape, and sharpness along the rim of a crater, are comparatively less examined. Quantifying these asymmetries helps better understand impact processes and surface evolution, thereby improving the interpretation of impact craters. Previous work on impact crater asymmetry has largely relied on qualitative descriptions, focusing on visual interpretations of rim morphology and ejecta patterns, whereas quantitative morphometric approaches have been employed only to a limited extent. This study aims to investigate whether the observed rim-crest asymmetry shows a systematic dependence on latitude, diameter, geological units, and/or age. Rim crest asymmetries can result from several factors, including the slope of the surrounding terrain, the angle of impact, and subsurface structural heterogeneities. This study is primarily restricted to fresh craters to ensure that the measured geomorphic parameters closely represent primary impact-generated features, minimize post-impact modification, and reflect intrinsic impact processes rather than secondary degradation. A quantitative evaluation of rim crest asymmetries was carried out for 1224 fresh lunar craters, with simple morphologies, ranging from 5 to 15 km in diameter, from the catalog of Povilaitis et al. (2018), using high-resolution (5 m/pixel) Digital Elevation Models (DEMs) generated from the Chandrayaan-2 Terrain Mapping Camera-2 (TMC-2). A Convolutional Neural Network (CNN) model specifically trained to distinguish fresh craters from degraded ones was used to filter out the sample craters. Furthermore, since Chandrayaan-2 does not provide a global lunar DEM mosaic, candidate craters were screened in the Chandrayaan-2 Data Explorer and map browser to select only those within the available DEM strips. For each fresh crater, rim crest maxima, minima, range, mean, standard deviation, spatial coordinates, and corresponding TMC strip product ID were recorded. This dataset constitutes the first global catalog of fresh, simple craters on the Moon, derived from Chandrayaan-2 TMC-2 DEM data and designed to facilitate future studies of lunar crater morphometry. Results indicate that rim-crest asymmetry is widespread. Even fresh craters often display uneven rim uplifts. Scatter plot analysis reveals that rim crest range does not correlate significantly with latitude ($R^2 < 0.01$), although the mean rim height shows a positive scaling relation with crater diameter ($H \propto D^n$, $R^2 = 0.2522$). The scaling exponent, $n = 1.0884$, exceeds the predictions of the classical simple crater model, suggesting that pre-impact surface selection substantially influences rim height. This inference highlights that rim-crest asymmetry is a common characteristic, independent of location, size, or terrain. Furthermore, correlations between patterns of asymmetry and external variables, such as regional topographic slope and post-impact modification processes (e.g., wall slumping or ejecta redistribution), will be investigated in future work.

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1. Introduction

Impact cratering is a fundamental geomorphological process on the Moon that shapes the lunar surface, with subsequent impacts continue to modify these features. Unlike Earth, where the surface is constantly reshaped by tectonic movements, erosion, and water, the Moon preserves impact craters, except for the subsequent impacts, and, therefore, creates a unique environment for studying impact cratering and their modification processes (Stöffler et al., 2006; Grier and McEwen, 2001; Mazrouei et al., 2019; Melosh, 1989; Pike, 1977). The rim crest is one of the most important morphological features of an impact crater, formed during the excavation and modification phase of crater evolution, and represents the highest topographic feature produced by impact-related uplift and collapse in a simple crater (Grieve and Garvin, 1984; Sharpton, 2014), wherein rim height refers to the height of this rim crest relative to the pre-impact surface.

Although hypervelocity impacts are expected to produce circular, radially symmetrical craters under ideal conditions (such as vertical impact angle, homogeneous and isotropic target material, flat pre-impact surface, uniform gravity, and absence of atmospheric effects and pre-existing structural heterogeneities) (Melosh, 1989), impact craters frequently deviate from perfect symmetry, exhibiting features such as elevated rims in specific directions, elliptical or tilted rim shapes, irregular ejecta distributions, and dislocated central peak position (Goeritz et al., 2009; Herrick and Forsberg-Taylor, 2003). The majority of impact craters form at non-vertical impact angles (Barlow and Bradley, 1990), and as the impact angle decreases, asymmetry becomes more pronounced. In fact, they can range from 15° to 60° relative to the horizontal plane (Ivanov, 2001). Due to non-vertical impacts, there will be asymmetry in the excavation cavity and in the distribution of resultant ejecta (Herrick and Forsberg-Taylor, 2003). As a result, there will be a notable difference in rim height between the uprange and downrange directions (Gault and Wedekind, 1978; Pierazzo and Melosh, 2000) (Fig. 1). The rim crest in the downrange direction will be higher than that in the uprange direction (Schultz, 1992; Anderson et al., 2003). Both experimental investigations (Baldwin, 2008; Wünnemann et al., 2016; Allibert et al., 2023) and numerical modeling (Lv et al., 2025; Caldwell et al., 2018; Ormö et al., 2022) have been carried out to study impact cratering processes. Experimental investigations elucidate processes such as shock-wave propagation, excavation flow, and ejecta emplacement, while numerical models further elucidate how subsurface heterogeneity, impact angle, and target layering

regulate crater shape, rim uplift, and asymmetry. Herrick and Forsberg-Taylor (2003) showed that noticeable crater elongation or ellipticity develops only when impacts occur at very low angles (generally $<10\text{--}15^\circ$ from the horizontal), which represent a relatively small fraction (less than 5%) of natural impact events. Unlike ellipticity, rim height asymmetry can occur even when the crater appears circular in plain view. However, with the increased availability of high-resolution topographical data, it has become apparent that many impact craters lack radially symmetric rims. The differences in rim height around the crater rim are described as rim crest asymmetry in this study.

The target slope at impact can influence rim crest elevations (Kreslavsky and Head, 2016). Over geological timescales, micrometeorite impacts, regolith redistribution, space weathering, and mass wasting can further alter rim topography, increasing or decreasing the degree of asymmetry depending on local conditions. Although rim crest asymmetry is potentially important, it has never been quantitatively analyzed for a representative sample of simple lunar craters using high-resolution digital terrain models. This could be due to the limited availability of high-resolution topographic data, challenges in consistently defining rim crests, and the lack of robust quantitative analysis methods in earlier studies. Most previous work has been case studies (cf. Pike, 1977; Sharpton, 2014). In this study, we conducted a quantitative analysis of rim crest asymmetry for lunar craters with diameters between 5 and 15 km (simple craters), selected from the Povilaitis et al. (2018) catalog. The craters in this catalog that fall within the high-resolution TMC-2 digital terrain models (DTMs) were selected for this study (Fig. 2), which counts to 1224 fresh craters. TMC-2 DTM was used to derive rim crest data, such as maximum, minimum, and mean elevations, which were systematically picked along crater boundaries, and statistical parameters of height variation were also calculated to quantify asymmetry. By comparing craters located within the mare region, which corresponds to the Procellarum KREEP Terrane (PKT) in the classification of Jolliff et al. (2000), with those within the Highlands Terrane across different latitudinal belts (0° to $\pm 60^\circ$), this study aims to assess whether rim crest asymmetry is indicative of primary impact dynamics or target characteristics. In this context, this work will contribute to a better understanding of crater morphology on airless planetary bodies, including decoupling impact mechanics from target properties, probing subsurface geology, refining models of planetary surface evolution, and advancing comparative planetology.

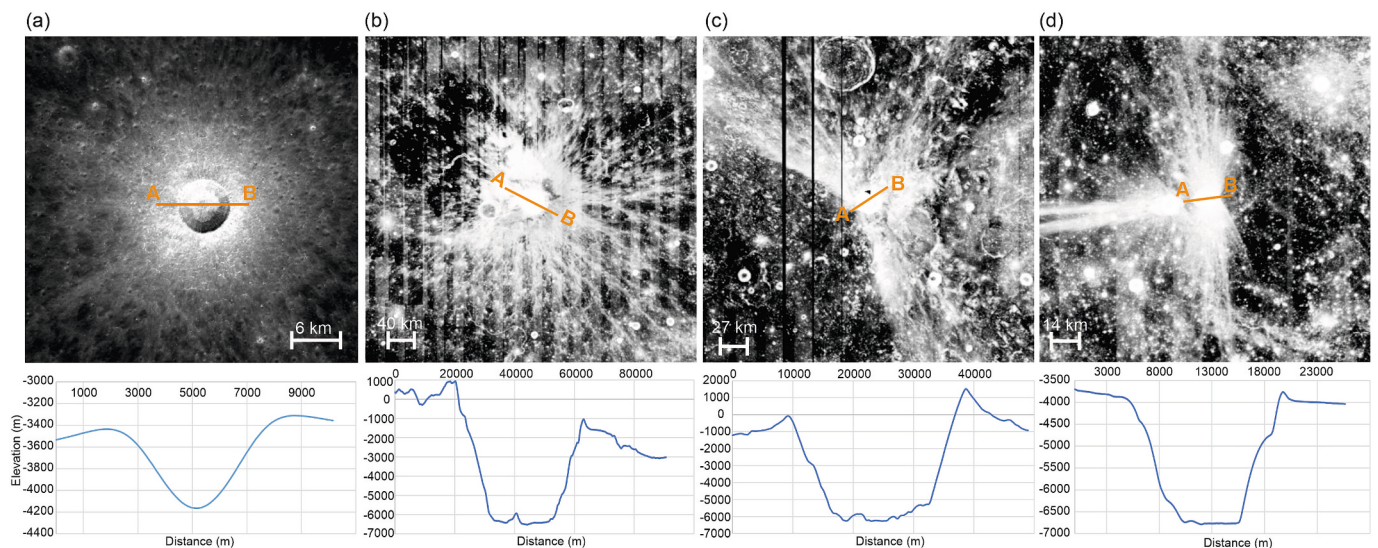


Fig. 1. Rim crest asymmetry and ejecta pattern variations as a function of impact direction, along with the corresponding cross profiles for (a) Moltke crater ($-0.59^\circ\text{S}, 24.16^\circ\text{E}$), (b) Aristarchus crater ($23.73^\circ\text{N}, 312.51^\circ\text{E}$), (c) Proclus crater ($16.09^\circ\text{N}, 46.89^\circ\text{E}$), and (d) Messier-Messier A crater ($-1.90^\circ\text{S}, 47.65^\circ\text{E}$).

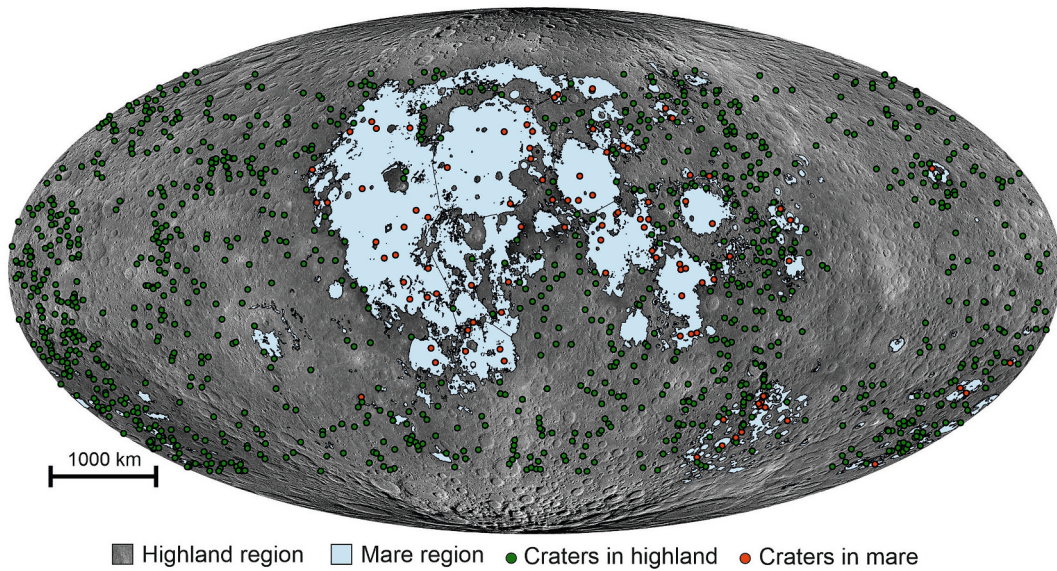


Fig. 2. Location map of selected craters across diverse geological and geomorphological environments.

2. Datasets and methods

The entire process of the methods adopted in this study is schematically shown in Fig. 3a,b.

2.1. Datasets used

We used the Povilaitis et al. (2018) lunar crater catalog, which provides a global inventory of craters ranging from 5 to 20 km in diameter and primarily includes simple morphology (which mostly constitutes bowl-shaped craters with well-defined rims and avoids the complexity of central peaks, terraces, or collapse features with slumped or degraded rims). The data can be downloaded from (https://data.lroc.im-ldi.com/lroc/rdr_product_select). Optical image patches for training a convolutional neural network (CNN) to classify fresh and degraded craters via supervised learning were generated from the Lunar Reconnaissance Orbiter Camera (LROC) Wide Angle Camera (WAC) global mosaic at 100 m/pixel resolution (Speyerer et al., 2011). This mosaic was constructed from more than 15,000 images acquired by the Lunar Reconnaissance Orbiter, which is freely accessible via the USGS Astrogeology portal (https://astrogeology.usgs.gov/search/map/moon/lro_lroc_wac_global_morphology_mosaic_100m). To further characterize the morphology of the identified fresh craters, topographic data from Chandrayaan-2 Terrain Mapping Camera (TMC)-2 DEM strips (5 m/pixel resolution) were employed. TMC-2 DEM data were chosen over LROC WAC because of their higher spatial and vertical resolution, which is vital for accurately extracting morphometric parameters such as rim height and subtle changes in the rim crest. Although WAC offers good global coverage, its lower resolution makes it less effective at recording fine topographical details, particularly craters between 5 and 15 km in diameter. The Chandrayaan-2 DTM strips can be downloaded from the Indian Space Science Data Center (ISSDC) PRADAN portal after creating a login ID (<https://pradan.issdc.gov.in/>).

2.2. Crater selection and pre-processing

The Povilaitis et al. (2018) database comprises 22,746 lunar craters. To avoid small secondary craters and large complex craters, the selected diameter range was 5–15 km, which encompasses only simple craters. The selected global crater inventory was partitioned into four longitudinal sectors (0–90°E, 90–180°E, 0–90°W, and 90–180°W), and all subsequent analyses were performed independently for each sector. Each crater was

represented by its geometric center using latitude and longitude attributes. ArcMap 10.8 was used to create crater rim polygons around each crater center, using each crater's diameter as input. The polygon was then converted into a polyline using the 'Feature to Line' tool in ArcMap, followed by the application of a 200 m buffer around the rim crest. A 200 m buffer was chosen for the 5 m/pixel DEM to include enough pixels (40) for reliable elevation measurements, reducing small, unwanted variations (noise) and uncertainty while still capturing local variations in the rim crest. Crater overlap checks were carried out using ArcGIS's 'intersect' tool to make sure each crater represented a unique impact event. Overlapped or nested craters can skew rim-elevation statistics and lead to inaccurate asymmetry measurements (Fig. 4a-i). Unlike global mosaics, such as the LROC WAC and the Lunar Orbiter Laser Altimeter (LOLA) DEM, the Chandrayaan-2 TMC-2 provides stereo-derived DEMs as individual strips. Each strip covers a narrow swath of the lunar surface, and when combined, they form a discontinuous dataset with substantial data volume and spatial segmentation. Initially, we tried to process multiple TMC-2 DEM strips in ArcGIS, which proved computationally intensive and inefficient. Therefore, to address these challenges, the data processing procedure was moved to Google Colab (<https://colab.research.google.com/>), a cloud-based platform with high computational efficiency and flexible data-handling capabilities, using Python libraries such as rasterio, geopandas, and rasterstats. Each TMC-2 DEM strip was analyzed directly within the Colab environment. This approach not only improved data-processing automation but also ensured that the rim crest elevation statistics were computed from the original high-resolution DEMs, without interpolation or resampling errors associated with mosaic generation.

2.3. CNN-based crater classification

2.3.1. Crater patch extraction

Crater image patches were extracted from the LROC WAC mosaic using crater coordinates and diameter from the Povilaitis et al. (2018) database. Also, a unique crater ID was assigned to each crater to maintain one-to-one correspondence between crater metadata and the extracted image patch. For each crater, a square window centered on the crater midpoint was cropped, with the window size scaled relative to crater diameter. Specifically, the patch width was defined as 2.5 times the crater diameter, ensuring inclusion of the rim, interior morphology, and immediate ejecta while minimizing background contamination. The resulting patches were saved and indexed using their corresponding crater IDs.

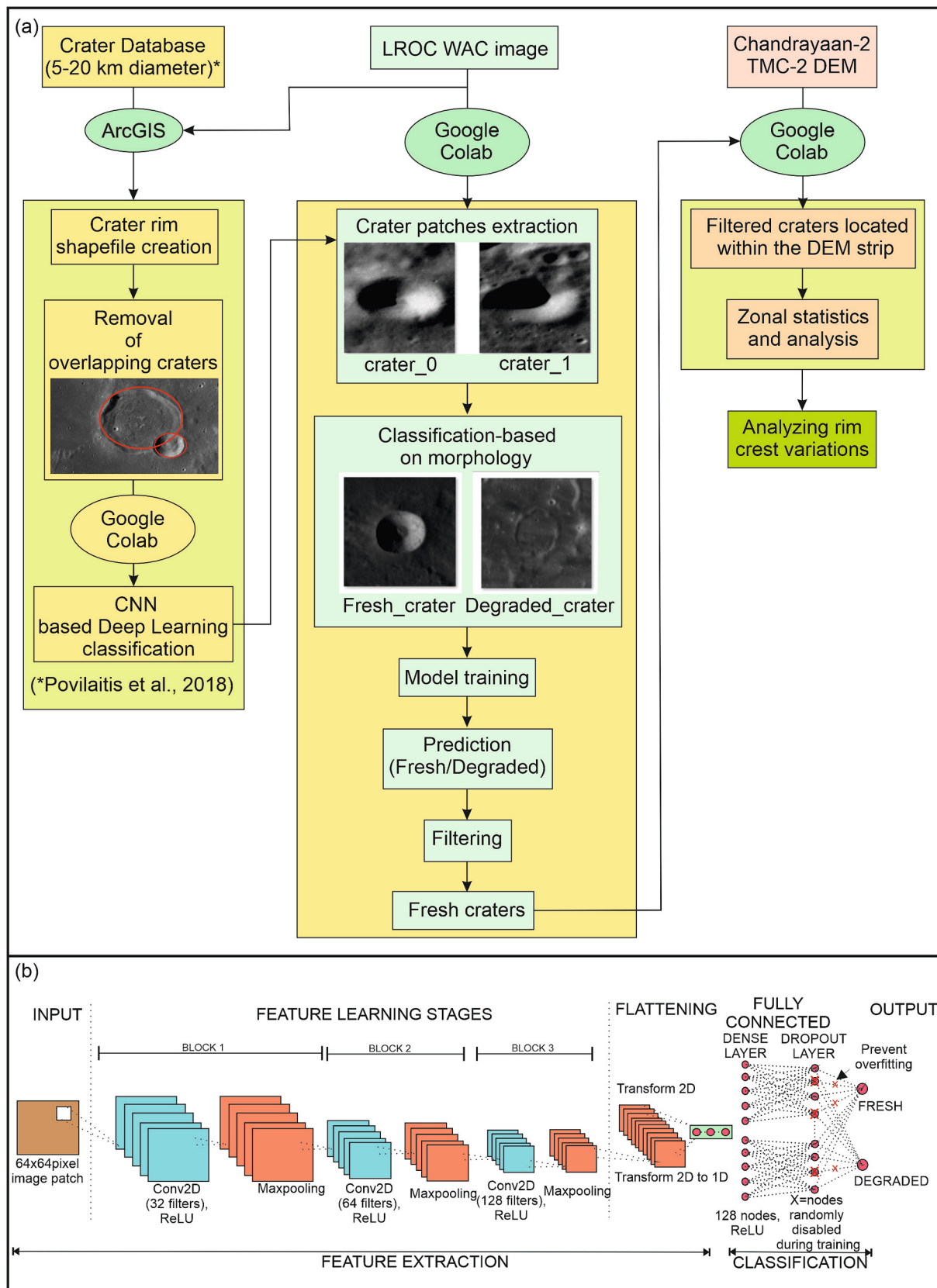


Fig. 3. (a) Flowchart illustrating the methodology adopted in this study (b) architecture of the Convolutional Neural Network (CNN) used for crater classification.

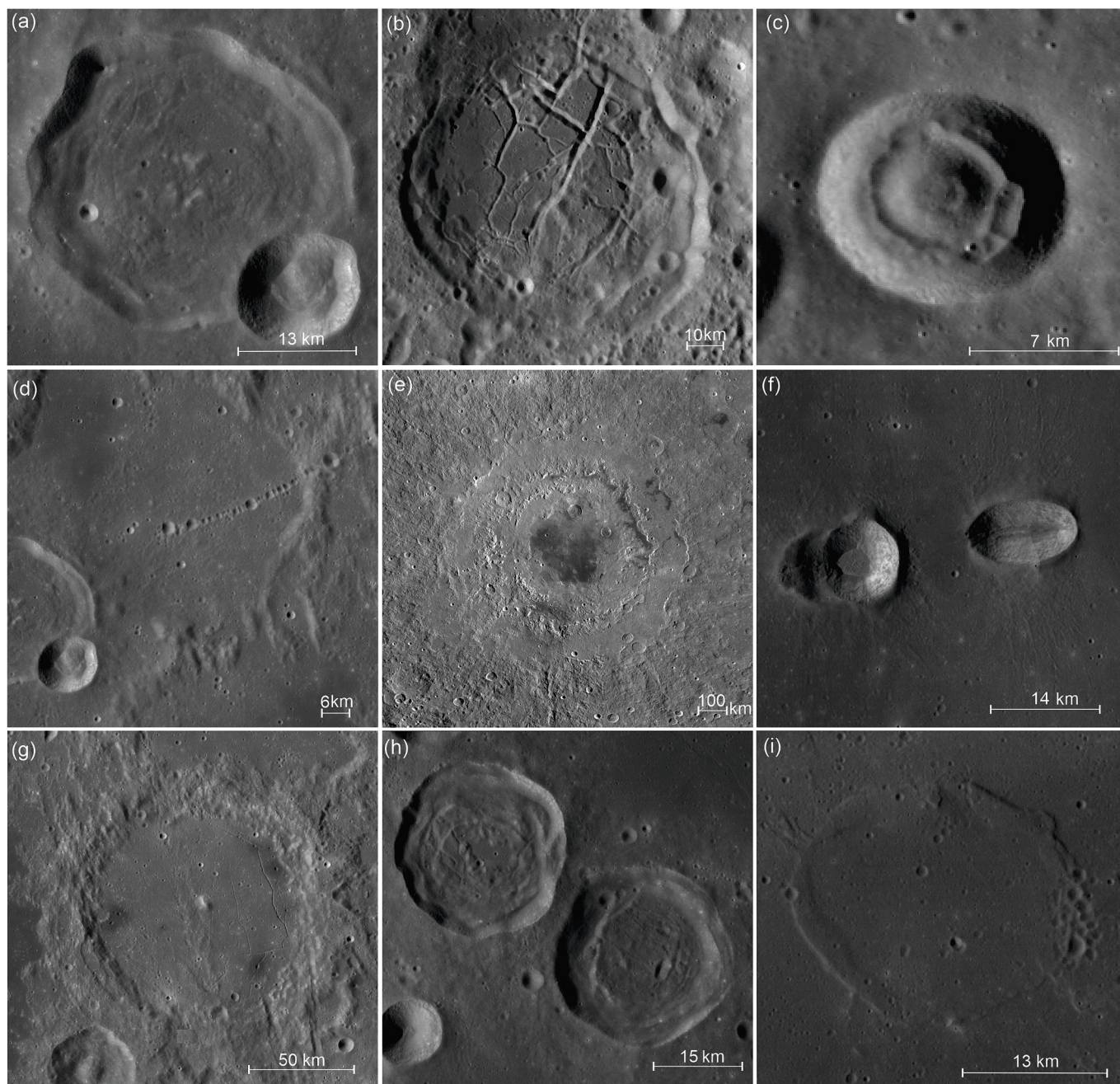


Fig. 4. Types of craters influencing rim crest asymmetry: (a) overlapping craters ($-11.85^{\circ}\text{S}, 351.85^{\circ}\text{E}$), (b) floor-fractured craters ($24.59^{\circ}\text{N}, 152.25^{\circ}\text{E}$), (c) concentric craters ($33.3^{\circ}\text{N}, 38.4^{\circ}\text{W}$), (d) catena craters ($11.0^{\circ}\text{S}, 7.0^{\circ}\text{W}$), (e) craters within large basins, (f) doublet craters ($-1.90^{\circ}\text{S}, 47.65^{\circ}\text{E}$), (g) nested craters ($13.4^{\circ}\text{S}, 2.85^{\circ}\text{W}$), (h) adjacent craters ($2^{\circ}\text{N}, 19^{\circ}\text{E}$), and (i) degraded (ghost) craters ($26.0^{\circ}\text{S}, 23.4^{\circ}\text{W}$).

2.3.2. Supervised learning framework and dataset preparation

After extracting crater patches (as mentioned in 2.3.1), a subset was visually inspected and manually labeled into two distinct morphological classes: (i) Fresh craters, characterized by sharp rims, visible ejecta patterns, and showing higher albedo contrasts to the surrounding, and (ii) Degraded craters, which exhibit ambiguous rims, infilled floors, and less distinct boundaries. An initial dataset comprising 125 fresh and 125 degraded lunar craters was manually labeled using established geomorphological criteria for crater freshness and degradation (Shao et al., 2025; Yang et al., 2020). These manually labeled datasets were used to create a supervised deep learning training set (Bickel et al., 2021; Silburt et al., 2019). To improve model generalization and address class imbalance, data augmentation was applied using random rotations ($\pm 30^{\circ}$), zooming, horizontal flipping, and spatial translations (in both horizontal and vertical

directions via width and height shifts). This was performed until each class contained approximately 400 images. The augmented dataset was subsequently split into training and validation sets in an 80:20 ratio, ensuring equal class representation in both sets.

2.3.3. CNN-based crater classification

CNNs are particularly well-suited for crater classification tasks because they automatically learn and extract features directly from imagery, surpassing conventional machine learning approaches. A CNN model was developed from scratch using TensorFlow and Keras, trained entirely on the prepared crater dataset without transfer learning or pre-trained models, to classify craters as fresh or degraded. TensorFlow and Keras were primarily used to efficiently build, train, and test deep learning models. Our model consisted of three convolutional blocks, each

comprising a convolution (Conv2D) layer, a ReLU activation, and a max-pooling layer, with 32, 64, and 128 filters, respectively. In CNNs' max-pooling layers, the spatial size of the feature maps was reduced while retaining only the essential features, thereby minimizing noise and computational complexity. These feature maps were fed into a flattening layer, which transformed the 2D features into a 1D representation; a fully connected dense layer with 128 nodes and a ReLU activation function was then used. Lastly, a dropout layer was added to prevent overfitting and improve the model's generalization by randomly disabling nodes at a rate of 0.5.

For model training, we used the Adam optimization algorithm, which dynamically adjusts the learning rate to improve efficiency and stability. In addition, we employed binary cross-entropy as the model's cost function. During model training, we applied a maximum of 30 epochs. Each epoch refers to a single pass over the entire dataset, allowing the model to learn feature representations iteratively. To prevent overfitting, training was performed with the EarlyStopping callback (patience = 4), which continuously monitored the validation loss and automatically terminated training after 4 consecutive epochs of no further improvement. Or else overfitting may occur if training is allowed to continue beyond a certain point. In addition, we used model checkpointing to select the model with the lowest validation loss. Model training was monitored using loss and accuracy curves. The highest-performing model achieved approximately 91% accuracy and was used to predict craters across the entire dataset.

2.3.4. Model prediction, classification, and filtering of fresh craters, and their integration with the TMC-2 DEM of Chandrayaan-2

The trained model was used to predict class probabilities for each crater image for the four longitudinal sections. To categorize images as either Fresh (≥ 0.5) or Degraded (< 0.5), a threshold value of 0.5 was selected. It is the standard decision boundary for binary classification problems using a sigmoid activation function, where the output represents the probability of belonging to the positive class. Values ≥ 0.5 indicate a higher likelihood of the "Fresh" class, while values < 0.5 correspond to the "Degraded" class. In this study, alternative threshold values were also explored to assess the sensitivity of the CNN-based crater classification (fresh/degraded), with 0.5 retained as the final threshold because it provided the most balanced separation between the two classes. The predicted class labels and associated probabilities were integrated with the original crater metadata and exported as a CSV file for further spatial and statistical analysis. Only that 0.39 fraction of craters from the main catalog was predicted as fresh by the CNN model, which were retained for rim-crest asymmetry analysis. This ensures that the elevation values are representative of primary morphology with minimal post-impact modification, allowing a more accurate assessment of original structural asymmetry that may have occurred due to several factors, such as oblique impacts (Poelchau and Kenkmann, 2008), variation in target material properties (different rock types and their strength, layering, and different regolith thickness) (Hörz and Cintala, 1997), and post-impact deformation. The filtered crater dataset was spatially intersected with the downloaded TMC-2 DEM strips of Chandrayaan-2. Only craters entirely within the DEM extent were selected for analysis, ensuring complete elevation data for each crater rim.

2.4. Statistical analysis

2.4.1. Extraction of rim crest elevation values

Using crater rim polygons (with a 200 m buffer around the crater rim crest) as a region of interest, zonal statistics were calculated to quantify topographic variations within individual lunar craters using Chandrayaan-2 DEM data. All corresponding DEM tiles (GeoTIFF format) were first checked for readability and consistency in coordinate reference systems. Raster spatial extents were then evaluated to identify DEM tiles intersecting each crater polygon, and zonal statistics were calculated only for overlapping raster crater pairs. For each crater, the following parameters were extracted from the corresponding raster strip on which the polygon

overlap: Minimum elevation ($E_{\min}$), Maximum elevation ($E_{\max}$), Mean elevation (E_{mean}), Standard deviation (σ_{elev} = how much the rim elevation differs from the mean rim elevation of that crater), and Elevation range ($\Delta E = E_{\max} - E_{\min}$). The process was automated in Python, with batch saving to ensure recovery during long computations. These parameters serve as a basis for quantifying asymmetry, which represents topographic variability along the crater rim. The resulting statistics were exported as a CSV file, forming the basis for subsequent analysis of rim-crest asymmetry and morphometric variability across the study region.

2.4.2. Analysis of rim crest asymmetry

To quantify asymmetry, one of the easiest ways is to measure the difference between the highest and lowest rim crest elevations and standard deviation along the crater rim. A low range or standard deviation indicates that the rim crest elevations are relatively uniform around the crater rim, i. e., symmetrical and evenly elevated around its circumference. Whereas large elevation differences around the rim signify asymmetry. Asymmetry analysis was also performed to compare variability in rim crest elevation across lunar latitudes (0–60°N and S) and to explore potential crustal effects. To assess this, the Range parameter was derived as

$$\text{Range} = h_{\max} - h_{\min} \quad (1)$$

where $h_{\max}$ represents the maximum rim crest elevation and $h_{\min}$ is the minimum rim crest elevation. This parameter effectively captures the spread of rim elevation values. Thus, the obtained rim-height range represents the vertical amplitude of asymmetry for each crater and serves as the basis for subsequent statistical analyses.

2.4.3. Rim height calculation

Rim height was quantified to evaluate vertical variations and asymmetry around each crater. The rim height (H_{rim}) was computed as the elevation difference between the crater rim and the mean pre-impact surface level according to the relation:

$$H_{\text{rim}} = Z_{\text{rim}} - Z_{\text{pre-impact}} \quad (2)$$

where Z_{rim} represents the maximum rim crest elevation value extracted along the crater perimeter, and $Z_{\text{pre-impact}}$ is the median elevation of the pre-impact terrain surrounding the crater. The pre-impact level was derived from the average of elevation points at a radial distance of 0.60D from the crater center, where D denotes the crater diameter. This threshold was selected based on visual assessment of the surrounding topography. To represent a region sufficiently beyond the crater rim and immediate ejecta deposits, where the topography is less influenced by impact-related modifications and better reflects the original surface conditions prior to impact. At distances closer to the rim, the terrain is significantly altered by processes such as rim uplift, ejecta emplacement, and wall slumping, which can distort the true pre-impact morphology (Melosh, 1989; Holsapple, 1993; Ivanov, 2001). The method provides an approximation of the pre-impact surface rather than an exact reconstruction. While regional slopes may persist, averaging elevation values beyond the crater rim minimizes local topographic variability and yields a consistent and representative baseline for morphometric analysis. This approach minimizes local slope effects and preserves representative pre-impact topography.

Given that the rim crest contains multiple elevation points, the following parameters were calculated to capture elevation variability:

$$H_{\max} = Z_{\text{rim}(\max)} - Z_{\text{pre}(\text{mean})} \quad (3)$$

$$H_{\min} = Z_{\text{rim}(\min)} - Z_{\text{pre}(\text{mean})} \quad (4)$$

$$\text{Rim height range} = H_{\max} - H_{\min} \quad (5)$$

Using the average of rim and pre-impact elevations will effectively reduce random topographic noise and local irregularities. Thus, the obtained rim-height range represents the vertical amplitude of asymmetry for each crater and serves as the basis for subsequent statistical analyses.

3. Results

3.1. CNN-based crater classification and data extraction

The CNN model achieved 88.91% training and 91.87% validation accuracy, with early stopping used to prevent overfitting. Figs. 5a,b show stable convergence between the training and validation datasets, indicating good generalization. The model successfully classified craters into fresh and degraded categories with high confidence scores (>0.8 for most predictions). About 5191 of the total craters were identified as fresh. For each CNN-classified crater, the corresponding Chandrayaan-2 TMC-2 DEM data were used to extract rim crest elevation values.

3.2. Terrain-based variability in rim crest elevation

To examine the influence of geological terrain on rim crest morphology, a comparative analysis was conducted between craters in mare and highland terranes. Specifically, the distribution of rim crest elevation range was evaluated to determine whether contrasting lithology, surface age, and crustal properties exert measurable control on rim crest asymmetry. As shown in Fig. 6, the violin-box plot illustrates the range of rim crest elevations for both mare and highland terrane craters. The box plot summarizes the dataset, showing the central tendency and interquartile range for each terrain type, along with the distribution of each. The violin plots show significant overlap in the distributions of rim crest range between mare and highland craters. Nevertheless, the median rim crest range is higher for highland craters (1363 m) than for mare craters (589 m), indicating a difference in their central tendencies despite the considerable overlap between the two distributions. Both populations exhibit positively skewed distributions, with a higher median and greater dispersion of rim crest range values in the highland population. To test whether this difference in dispersion could be attributed solely to the unequal sample sizes (101 mare and 1123 highland craters), we performed a Monte Carlo matched-sample subsampling analysis. In each of 1000 iterations, 101 highland craters were randomly selected without replacement, matching the sample size of the mare population, and the standard deviation (SD) and interquartile range (IQR) of the rim crest range (ΔH) were calculated. The median SD and IQR of the highland subsamples were 1005.69 m and 1236 m, respectively, compared with 332.35 m and 298 m for the mare population. The 95% empirical intervals for the highland SD (826.18–1190.13 m) and IQR (925.98–1554.10 m) were also substantially higher than the corresponding mare values. None of the 1000 matched highland subsamples produced an SD or IQR equal to or lower than the mare value (Supplementary Material 1). These results indicate that the greater dispersion observed in the highland population is not simply due to its larger sample size. Additionally, to assess whether the differences in the

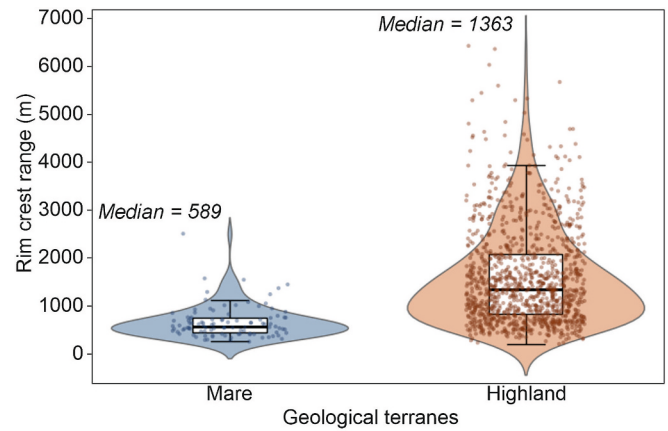


Fig. 6. Violin-box plots showing the distribution of rim crest range for mare and highland craters.

rim crest range between mare and highland craters are significant, a non-parametric Mann–Whitney U test was applied (Feltovich, 2003). The data are not normally distributed and exhibit skewness, which makes the Mann–Whitney U test more appropriate than other parametric tests. The Mann–Whitney U test compares the distributions of two independent groups and tests whether one group has systematically higher or lower values than the other. This test was conducted to determine whether the geological setting significantly affects the elevation range of the rim crest.

The test yielded a U statistic of 16,840.0 with a p -value of 1.04×10^{-31} , indicating that the distributions of rim crest range between the two geological terrains are statistically different ($p < 0.001$). The median rim crest range is 589 m for mare craters and 1363 m for highland craters, indicating that highland terrains exhibit significantly greater rim crest variability in highland craters compared to mare craters. This difference is consistent with the contrasting geological characteristics of the two terranes, as the heavily cratered and lithologically heterogeneous highlands are expected to produce greater morphometric variability than the relatively smooth basaltic maria (Melosh, 1989; Pike, 1977; Lalor and Sharpton, 2014). The standardized Z value is -11.72 , and the calculated effect size is very large in magnitude ($r = 0.335$). According to Cohen's (2013) definition, the effect size is negligible ($r < 0.1$), small ($0.1 \leq r < 0.3$), moderate ($0.3 \leq r < 0.5$), and large ($r \geq 0.5$). The obtained effect size ($r = 0.335$) is moderate, indicating a meaningful difference between the two groups. Therefore, the effect of geological terrain is meaningful for rim crest variability.

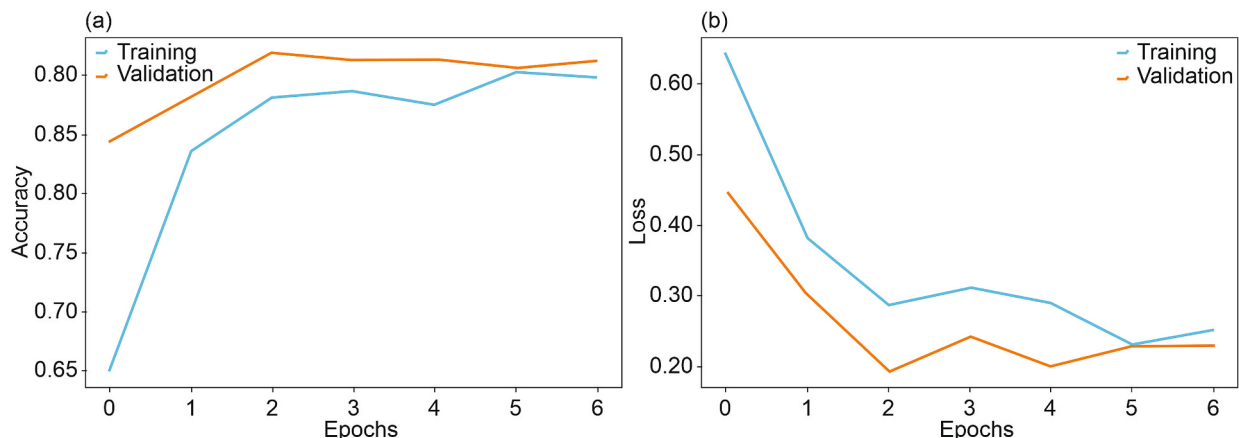


Fig. 5. Training history of the CNN model showing (a) training and validation accuracy and (b) training and validation loss as a function of epoch.

3.3. Influence of latitude on rim crest asymmetry

The potential impact of geographic location was also analyzed by examining the relationship between latitude and the range of rim crest elevations. The relationship between latitude and the range of rim crest heights shows no meaningful correlation across the analyzed crater population (Supplementary Material 2). Here, the linear regression yields a best-fit equation of $y = -0.3382x + 1531.6$, with a coefficient of determination (R^2) = 0.0001. The very low R^2 value indicates that latitude accounts for less than 0.01% of the variability in rim crest range. The regression slope is slightly negative; however, the scatter plot shows significant variability in the rim crest range values at all latitudes between 60°S and 60°N. High-variability outliers were observed at all latitudes, and there is no indication of a systematic increase or decrease in the asymmetry pattern towards the equator or the mid-latitudes. The regression trends were nearly horizontal, suggesting that rim crest variation is independent of craters' latitudinal location.

3.4. Diameter scaling and rim crest variability

Further extension of the above analysis aims to examine the possible influence of crater size on rim crest asymmetry by assessing the relationship between crater diameter and rim crest range. A scatterplot of diameter and range distribution was also plotted (Fig. 7). The regression analysis yielded a coefficient of determination ($R^2 = 0.1508$) and a weak positive relationship between diameter and rim crest range ($y = 154.31x + 297.94$), indicating that range tends to increase slightly with increasing diameter, although substantial variability exists across all diameters.

Similarly, the analysis of rim crest heights indicates a positive trend between average rim height and diameter, consistent with well-established scaling laws for impact craters (Pike, 1977; Housen et al., 1983). The average rim height is observed to increase with diameter, and this is consistent with a power-law trend, given by $H_{\text{mean}} \propto D^n$, where n is a power-law index (Pike, 1977) (Fig. 8a). Linear regression between average rim height and diameter shows a moderate positive correlation ($R^2 = 0.253$), indicating that average rim height increases with diameter, though with considerable scatter, likely due to target heterogeneity, degradation, and post-impact modification. Log-log regression is used to further clarify the scaling trend (Fig. 8b), yielding a comparable correlation ($R^2 = 0.2522$), confirming the power-law relationship between the average rim height and diameter. Overall, the results indicate that the rim crest morphology is consistent with the scaling laws, although the scatter and asymmetry are likely due to secondary geological factors. When analyzed separately by terrain type (Fig. 8c,d), mare craters exhibit a stronger scaling relationship between average rim height and diameter compared to highland craters. The log-log regression for mare craters yields a higher R^2 (0.4089), indicating a more consistent power-law relationship between rim height and crater diameter. In contrast, highland craters show a weaker correlation ($R^2 = 0.2402$). Notably, the derived

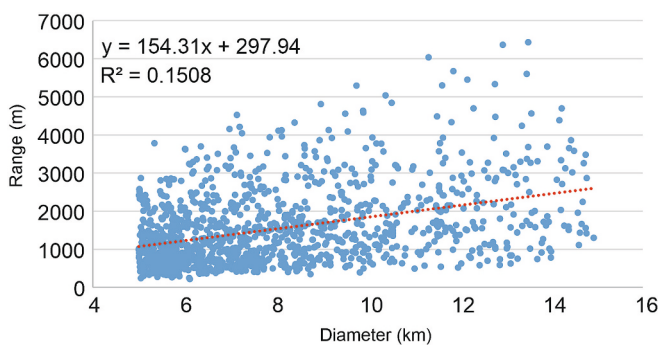


Fig. 7. Scatter plot showing the relationship between crater diameter and rim crest range.

scaling exponent for mare craters ($n = 1.0588$) exceeds classical simple-crater theoretical predictions. However, when comparing the mare and highland populations using the Mann-Whitney U test, the results show a very large effect size ($r = -0.335$), suggesting that surface lithology and target properties also play a significant role in controlling rim crest asymmetry.

Additionally, to assess the potential influence of crater diameter on rim-crest asymmetry, the rim-crest range was normalized by crater diameter ($\Delta H/D$) (Supplementary Material 3). Unlike the absolute rim-crest range, the normalized rim-crest range showed virtually no relationship with crater diameter ($R^2 = 0.0019$), indicating that the apparent increase in absolute rim-crest range with crater diameter is largely reduced after normalization. The normalized analysis yielded trends consistent with those obtained using the absolute rim-crest range. The relationship between terrain surface age and normalized rim-crest asymmetry remained weak and statistically insignificant (Spearman $\rho = -0.230$, $p = 0.116$). These results indicate that the observed age-related pattern is robust to crater-size normalization and is not substantially controlled by variations in crater diameter (Supplementary Material 4). Thus, statistical analysis of 1224 fresh lunar craters shows that rim-crest elevations exhibit considerable variability. The range between maximum and minimum rim crest elevations consistently exceeds the standard deviation of rim heights, indicating that deviations from uniform rim morphology are common across the crater population.

To further assess the reliability of the Chandrayaan-2 TMC-2-derived topographic measurements, the availability and spatial coverage of higher-resolution Lunar Reconnaissance Orbiter Narrow Angle Camera (NAC) DTMs were also examined for the 1224 craters included in the final dataset. However, none of the selected craters are covered by available NAC DTMs, precluding a direct crater-by-crater comparison. Therefore, the reliability of the TMC-2-derived morphometric measurements was instead assessed by considering the reported vertical uncertainty of the DEM and its propagation into the derived rim crest range (Amitabh et al., 2020). Using the reported vertical RMS error of ± 25 m for the maximum and minimum rim crest elevations, the uncertainty in the rim crest range was calculated using standard error propagation:

$$\sigma\Delta H = \sqrt{(\sigma H_{\text{max}})^2 + (\sigma H_{\text{min}})^2} \quad (6)$$

where H_{max} is the maximum rim crest elevation, H_{min} is the minimum rim crest elevation, and ΔH is the rim crest range, yielding an uncertainty of approximately ± 35.4 m. The resulting uncertainty is small relative to the observed rim crest ranges, supporting the reliability of the derived measurements.

4. Discussion

4.1. Prevalence of rim crest asymmetry and its scaling behavior

This study shows that the asymmetry in the rim crest is not an isolated geomorphological feature but rather a significant, widespread property of lunar craters. Indeed, most of the analyzed craters show measurable asymmetry in rim crest elevations between the maximum and minimum values, and the relative values of rim crest asymmetry clearly indicate that the proportion of completely symmetric craters is negligible compared to that showing considerable asymmetry. This result indicates that, although ideal circularity and radial symmetry provide useful first-order approximations for vertically formed simple craters in analytical and numerical models of craters (Elbeshausen et al., 2013), actual crater rims exhibit measurable topographic variability along their circumference. These elevation variations along the rim can be quantitatively described by measurable topographic heterogeneity resulting from crater formation and subsequent evolution (Wünnemann, 2010). The statistical values (range and standard deviation) across different diameter classes show only modest variation; however, the observed positive trend and low coefficient of determination ($R^2 = 0.15$)

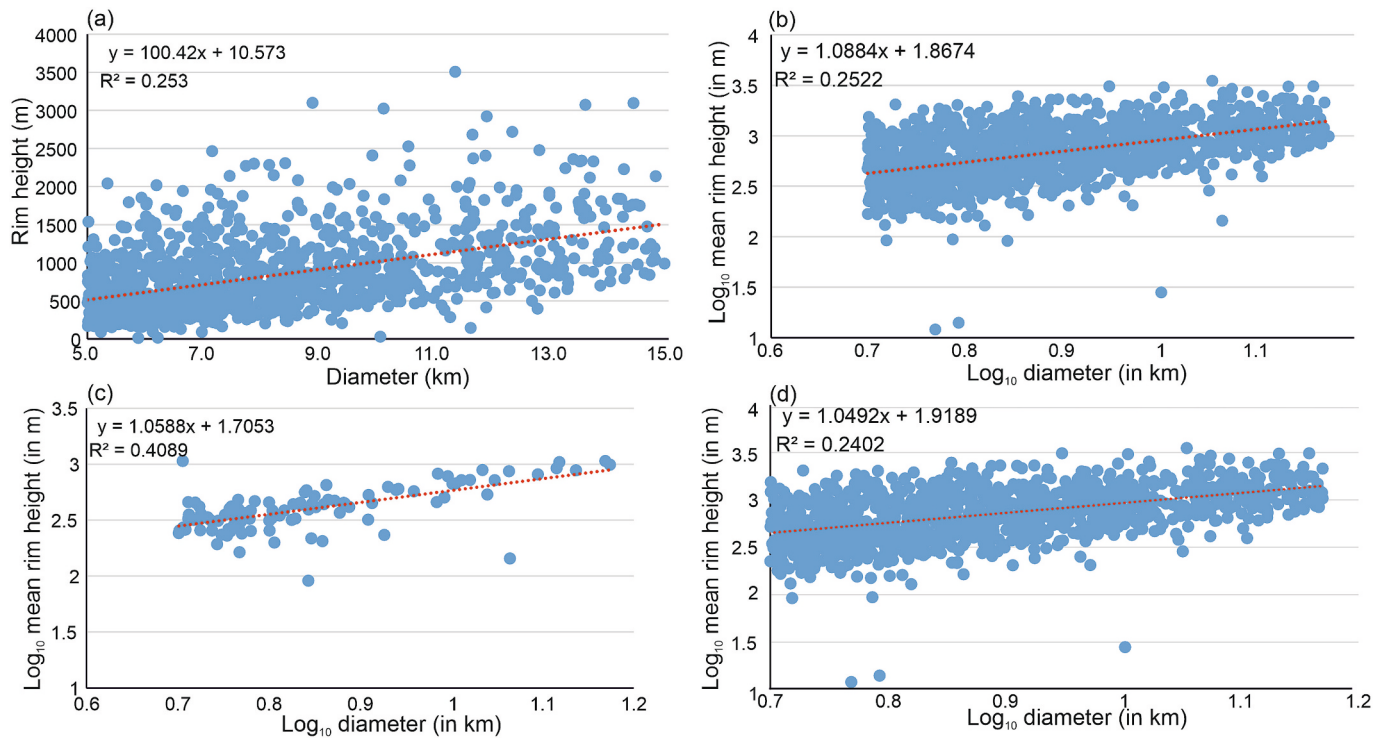


Fig. 8. (a) Scatter plot showing the relationship between crater diameter and average rim height; (b) logarithmic plot of diameter versus mean rim height; (c) log–log regression plot for mare craters; and (d) log–log regression plot for highland craters.

indicate that rim crest asymmetry is weakly dependent on crater size. The log–log relationship of the diameter of the craters with the range of the average rim height conforms to a power-law relationship, as expected by the classical simple crater scaling relations of Pike (1977). The diameter–rim height relationship remains valid at the first-order level, as average rim height increases systematically with crater diameter. However, analysis of maximum and minimum rim crest elevations reveals considerable circumferential variability that is not captured by average-based scaling. These deviations indicate that classical diameter–height relations adequately describe bulk crater geometry but fail to represent local morphometric heterogeneity along the rim crest. Rim crest asymmetry thus represents a second-order geometric variability superimposed on the primary scaling trend. Consequently, relying solely on average rim height obscures the internal elevation and yields an incomplete representation of crater morphology; thus, incorporating the rim crest elevation range provides a more rigorous quantification of geomorphic complexity.

4.2. Controls, formation, and implications of rim crest asymmetry

One of the major controls on rim crest asymmetry appears to be target heterogeneity at the surface and subsurface levels, including differences in regolith thickness, fracture density, and compositional differences between mare and highland, all of which influence impact excavation and ejecta emplacement. In heterogeneous targets, variations in material properties can influence shock-wave propagation, excavation, and rim uplift, potentially contributing to variations in the final crater morphology (Melosh, 1989; Wünnemann et al., 2006; Collins et al., 2008). Although such heterogeneity may affect ejecta distribution and other aspects of crater morphology, it does not necessarily produce an asymmetric rim crest. The greater variability in rim crest asymmetry observed across certain terrain types may therefore reflect the combined influence of lithological differences and other factors affecting crater formation and modification. Differences in target strength and pre-existing structural weaknesses can influence excavation and rim uplift,

collapse dynamics, and the final crater geometry (Melosh, 1989). In addition, pre-impact topography may play a significant role in controlling rim asymmetry. Impact craters formed on inclined or regionally sloping terrain tend to exhibit rim uplift on one side and increased ejecta thickness on the downslope side, suggesting that even slight slopes can influence rim height differences and contribute to the observed asymmetry, rather than it being solely due to impact processes (Elbeshausen et al., 2013; Omura et al., 2021; Aschauer and Kenkmann, 2017; Krohn et al., 2021). Even though overlapping craters were excluded from the dataset, degradation mechanisms such as seismic activity and ejecta blanketing cannot be entirely avoided. However, the persistence of asymmetry among relatively fresh craters suggests that rim crest asymmetry is more closely related to formation processes, with degradation likely enhancing pre-existing asymmetry rather than generating it independently. Terrain-based comparisons also underscore the importance of target characteristics, with increased asymmetry dispersion in the highland craters possibly related to greater crustal variability and fracture complexity, whereas the more homogeneous mechanical properties of the mare basalts may yield lower variability. The widespread occurrence of rim crest asymmetry implies that transient cavity collapse is not perfectly axisymmetric, rim uplift is spatially variable, and the distribution of impact energy interacts strongly with target properties, thereby challenging simplified analytical models that assume ideal symmetry and highlighting the need to incorporate realistic terrain parameters into numerical simulations.

4.3. Challenges and limitations

The present study is subject to several data-processing-related and computational limitations. The unavailability of a global Chandrayaan-2 TMC-2 DTM mosaic limits global lunar crater analysis, and existing datasets contain spatial data gaps that hinder continuous crater rim mapping. Thus, craters located within the available DEM extents and within a specific crater diameter range were considered for analysis. The analysis was carried out in four longitudinal sections to ensure systematic spatial

coverage and to manage computational constraints. Processing the entire dataset at once exceeded memory limits and reduced computational efficiency, particularly during high-resolution DEM processing and CNN-based classification. Segmenting the study area into longitudinal sections allowed for optimized processing, reduced memory load, and improved workflow stability without compromising analytical accuracy. In addition, the analysis was restricted to 60°S–60°N, to minimize potential uncertainties in photogrammetrically derived Chandrayaan-2 TMC-2 DEM measurements (can be affected by factors such as image geometry, image matching, illumination, and surface characteristics) at extreme latitudes, excludes craters within the South Pole–Aitken Terrane (SPAT); consequently, the analysis is limited to comparisons between the mare and the highlands terrane.

Another area where the model could be improved is by increasing the size of the manually annotated dataset for supervised learning. This is due to time and other resource constraints, which led to only 125 craters being manually annotated for the initial training dataset. To some degree, manually identifying crater degradation depends on expertise, which adds to the subjectivity of the training dataset. While the present model delivered good classification results, increasing the size and diversity of the manual dataset could further improve the model. Although this study focuses exclusively on fresh craters to minimize the effects of post-impact modification, minor degradation cannot be entirely ruled out. Processes such as localized mass wasting (slumping or landslides), limited regolith accumulation, secondary impacts, and ejecta emplacement from nearby impacts may still subtly modify the rim morphology. These processes could partially obscure the original impact-related asymmetry and introduce a small degree of uncertainty into the asymmetry measurements.

5. Conclusion

This study is the first to demonstrate a robust methodology for detecting and quantifying rim-crest asymmetry in lunar craters using Chandrayaan-2 TMC-2 DEMs. The analysis of latitude vs rim crest range shows no meaningful correlation, as evidenced by extremely low coefficients of determination ($R^2 < 0.01$). In contrast, average rim height scales positively with crater diameter, broadly consistent with established impact scaling laws ($H \propto D^n$), although with moderate scatter ($R^2 = 0.2522$). Terrain-wise analysis shows stronger scaling coherence in mare craters compared to highlands. Overall, the results show that rim crest asymmetry is a widespread morphometric characteristic of simple lunar craters. While rim height follows expected impact-scaling trends, the magnitude of asymmetry shows no significant dependence on latitude, crater diameter, or major geological terrain, suggesting that intrinsic impact dynamics and local heterogeneity dominate the control on rim crest morphology. This implies that factors such as impact angle, velocity, energy distribution, local differences in target materials, and post-impact changes largely shape rim crest features. Despite the study's many strengths, it has some limitations. The analysis is confined to a specific diameter range and relies on TMC-2 DEMs that lack a global mosaic, potentially affecting the precision of rim crest detection. Also, excluding highly degraded or overlapping craters could limit the broader applicability of the findings to all crater conditions. Future research will expand the dataset to cover a wider range of crater sizes and preservation states and will also use global topographic datasets to enhance accuracy. Adding information about subsurface structures and material properties will help elucidate how variations in target properties influence the impact process. Numerical modeling and experimental impact studies will isolate the effects of impact angle and target characteristics on rim asymmetry. Applying this method to terrestrial impact sites will provide useful comparative insights and strengthen the understanding of lunar results.

CRedit authorship contribution statement

Arya Nandakumar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **A. Rajaneesh:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **M.B. Saagara:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Sachana Sathyan:** Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **S. James:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Conceptualization. **K.S. Sajinkumar:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.icarus.2026.117329>.

Data availability

The dataset of 1224 fresh craters analyzed in this study and the CNN model for fresh/degraded crater classification can be accessed and run in Google Colab, with the corresponding code and dataset files available on GitHub (https://github.com/AryaNandan/Lunar_crater_rim_asymmetry).

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